Chapter 9: Simple Linear Regression

Statistics for Environmental Science

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> *Essentially, all models are wrong, but some are useful.*
>
> George E. P. Box

Back in Chapter 2, we used the covariance between two variables to quantify the strength and direction of a relationship (Pearson's r), fit a best-fit line through a scatterplot, and judged r against a simulated chance window. What we did not yet have was a formal test of the question every other test in this book has asked:

Could chance alone plausibly have produced our result?

Now that we've learned hypothesis testing, sampling distributions, and the logic of comparing an observed statistic to a null distribution, we can apply that same logic to the regression line and test whether a given slope is evidence of a real relationship.

Regression basics

The equation for a linear regression will be familiar, and it's one of the most heavily used equations in applied science:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Here β_1 is the slope and β_0 is the intercept. These subscripted βs are regression coefficients and have no connection to β, the Type II error rate from Chapter 5.

A linear equation, fit to real data, is a statistical **model**. We never observe β_0 or β_1 directly. Real data also do not fall exactly on a line, so the model needs one more term, the **error term** ε, the variability in the DV that the population line doesn't explain.

Computing the best-fit line by hand

In practice, we let software identify the line with the smallest total squared error, but working through one example by hand shows where that line actually comes from. Here are the formulas for the slope and intercept, calculated straight from the data:

$$\text{intercept} = b_0 = \frac{\sum y \sum x^2 - \sum x \sum xy}{n \sum x^2 - (\sum x)^2}$$

$$\text{slope} = b_1 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

Here x and y are the individual scores. This is the same line from Figure 2.10 in Chapter 2. To find its slope and intercept, we start with this table:

Table 1: Sums for the slope and intercept formulas

Pair	x	y	x^2	xy
1	1	2	1	2
2	4	5	16	20
3	3	1	9	3
4	6	8	36	48
5	5	6	25	30
6	7	8	49	56
7	8	9	64	72
Sums	34	39	200	231

The table has 7 pairs of x and y scores, plus columns for x^2 and xy (each x score times its matching y score), with column sums at the bottom. These sums are everything the formulas need:

$$\text{intercept} = b_0 = \frac{\sum y \sum x^2 - \sum x \sum xy}{n\sum x^2 - (\sum x)^2} = \frac{39 \times 200 - 34 \times 231}{7 \times 200 - 34^2} = -0.221$$

$$\text{slope} = b_1 = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} = \frac{7 \times 231 - 34 \times 39}{7 \times 200 - 34^2} = 1.19$$

With b_0 and b_1 in hand, we can compute a residual for any observed point. A residual, as in Chapter 2, is the difference between what we observed and what the line predicts, $e = y - \hat{y}$, where $\hat{y} = b_0 + b_1 x$. For the first row above ($x = 1, y = 2$): $\hat{y} = -0.221 + 1.19(1) = 0.969$, so $e = 2 - 0.969 = 1.031$. That point is above the line. Repeating this for every point, then squaring and summing the residuals, gives the quantity that b_0 and b_1 were chosen to minimize.

The residual e is the sample's version of the error ε. The error ε belongs to the population line, which we never see, and the residual e belongs to the line we fit.

Parameters and sample statistics

The beta coefficients in the model are population parameters. They describe the population line, which we infer from a sample but never know for certain.

Symbol	What is it?	Do we know it?
β_0, β_1	True population intercept and slope	Almost never known for sure
b_0, b_1	Sample intercept and slope	Yes, calculated by fitting the line to our data

Worked regression example: stream temperature and dissolved oxygen

Warmer water holds less dissolved gas, so ecologists expect dissolved oxygen (DO) to decline as stream temperature rises. This matters for stream health, because low DO can stress or kill fish and invertebrates. Figure 1 lays out the design.

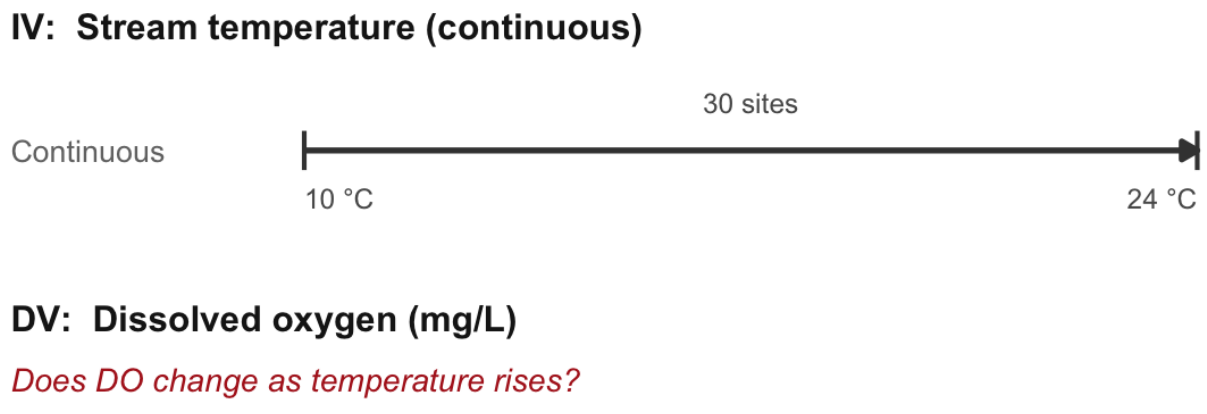


Figure 1: A regression design. Instead of a few groups to sort sites into, every site sits somewhere along a continuous range, and every temperature value is treated as its own group. A slope describes how the mean changes across that range.

Up to now every design put units into groups and asked whether the groups differed. Now, with a continuous predictor, every temperature value is treated as its own group. Each of the 30 sites falls somewhere along the range, and the line describes how mean DO changes from one temperature to the next. The question becomes how much DO changes per degree rather than whether two groups differ.

The 30 sites in this example are invented. We built them from a population line with $\beta_0 = 13.5$ mg/L and $\beta_1 = -0.28$ mg/L per °C, plus random scatter, so here, unlike in a real study, we know the true slope that b_1 estimates.

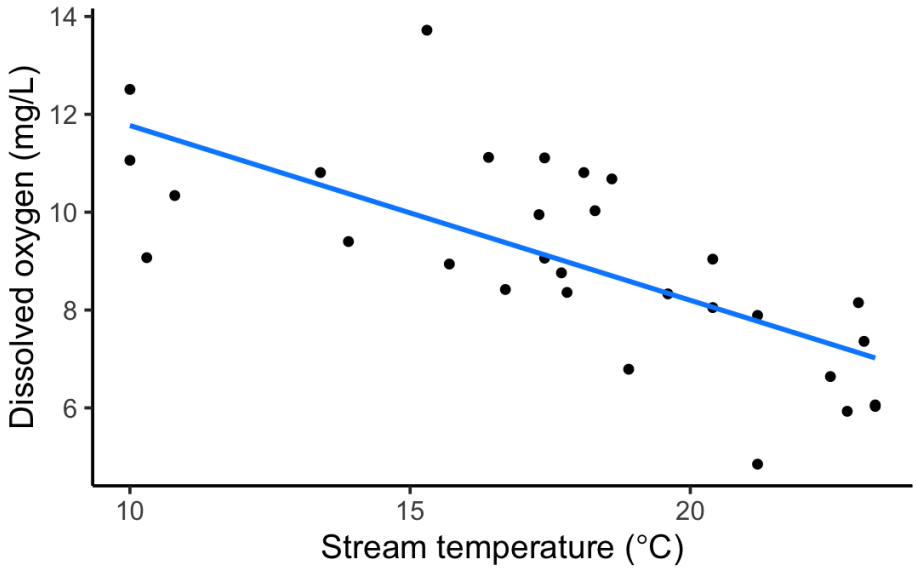


Figure 2: Stream temperature and dissolved oxygen at 30 sample sites, with the fitted regression line.

The scatterplot in Figure 2 shows a clear downward trend. Just from the graph, we can guess a couple of things:

- Pearson's r should be negative, since DO falls as temperature rises.
- It should be well short of -1, since the points scatter noticeably around the line.

Regression tests whether a relationship like this one is more than chance by writing it as a model.

The regression model

We model dissolved oxygen as a linear function of stream temperature, plus error:

$$DO = \beta_0 + \beta_1(\text{Temperature}) + \varepsilon$$

β_0 is the population intercept, β_1 is the population slope, and ε is the error, the part of DO that temperature alone doesn't explain.

We don't know β_0 and β_1. We estimate them from the sample as b_0 and b_1, using the same least-squares logic from Chapter 2 (the line that minimizes the sum of squared residuals).

In R, this is a single line of code:

```
fit <- lm(DO ~ stream_temp, data = stream_df)
```

This fitted line (Figure 3) is the best-fit line that minimizes the total squared distance between the data and the line.

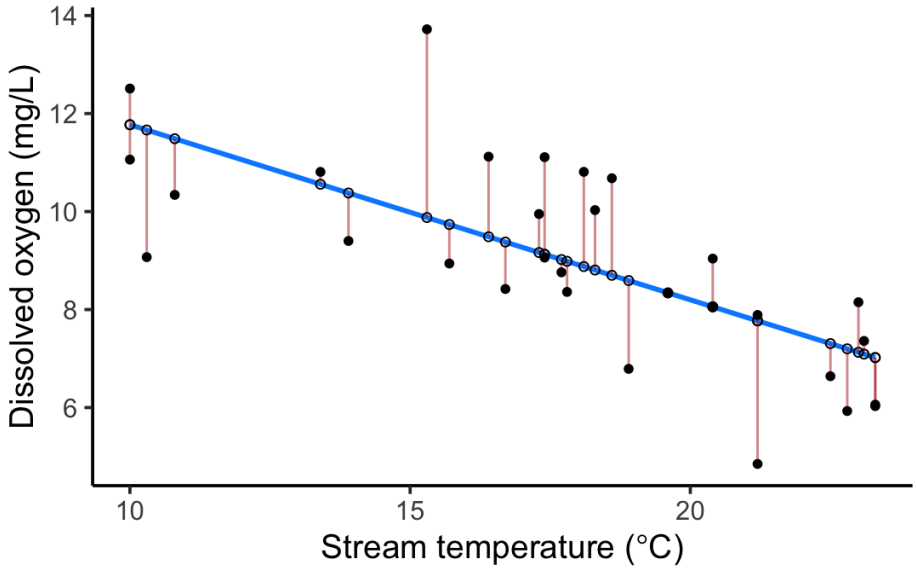


Figure 3: Stream temperature and dissolved oxygen with the fitted line. Vertical red segments are residuals.

`summary(fit)` prints the estimate, standard error, t, and p for each coefficient, and `confint(fit)` gives their 95% confidence intervals. The table below combines the two.

```
summary(fit)
confint(fit)
```

Table 2: Coefficient table for DO ~ stream temperature

Term	Estimate	SE	t	p	CI low	CI high
(Intercept)	15.34	1.23	12.47	< 0.001	12.82	17.87
stream_temp	-0.36	0.07	-5.30	< 0.001	-0.50	-0.22

With the results of the regression, we can get an equation for the fitted line:

$\widehat{DO} = 15.34 - 0.36 \times \text{Temperature}$.

The hat over DO marks it as a *predicted* value, what the model outputs for a given temperature, as distinct from an actually observed DO measurement.

In the coefficient table, the `Estimate` values for the intercept and slope are b_0 and b_1, not β_0 and β_1.

Table 3 says what each column of the coefficient table holds.

Table 3: Columns of the coefficient table, with the names R uses for them.

In this book	In R's `summary()` output	What it is
Estimate	`Estimate`	b_0 and b_1, the fitted intercept and slope
SE	`Std. Error`	How much that estimate would vary from sample to sample
t	`t value`	The estimate divided by its SE
p	Pr(>|t|)	The probability of a t at least this far from 0, in either tail, if the population coefficient were 0
CI low, CI high	`2.5 %` and `97.5 %`, from `confint()`	The bounds of the 95% confidence interval for that coefficient

The intercept ($b_0 = 15.34$) is the model's predicted DO at 0°C. That temperature is outside the range we sampled (10.0 to 23.3°C), so we won't interpret the intercept.

The slope ($b_1 = -0.36$) is the part we care about most. For every 1°C increase in stream temperature, predicted dissolved oxygen drops by about 0.36 mg/L.

Testing the slope

The slope b_1 is a sample estimate. Recall the hypothesis test asks whether the population slope β_1 could plausibly be zero:

- $H_0 : \beta_1 = 0$ (no linear relationship between stream temperature and dissolved oxygen)
- $H_a : \beta_1 \neq 0$ (there is a linear relationship)

Like any other statistic, b_1 varies from sample to sample. If we drew a new sample of streams and fit a new line, we'd get a slightly different b_1, just by chance. The test asks whether our b_1 is large relative to that sampling variability, the same signal-over-noise pattern as every test so far:

$$\text{statistic} = \frac{\text{effect}}{\text{error}}.$$

R reports the t-test for the slope in the coefficient table from `summary(fit)`, so there is no separate test to run. You get the `Std. Error`, t, and p for each coefficient at once.

The test statistic divides the estimated slope by its standard error, the SE column in the coefficient table above. The standard error measures how much b_1 would vary from sample to sample if we repeated this study many times:

$$t = \frac{b_1}{SE(b_1)} = \frac{-0.357}{0.067} = -5.30$$

with $df = n - 2 = 28$ (one degree of freedom spent estimating the intercept, one for the slope).

The p-value comes from a t-distribution. If temperature and DO were truly unrelated in the population, repeating this study many times would still produce a nonzero b_1 just by chance, and a t-statistic that follows

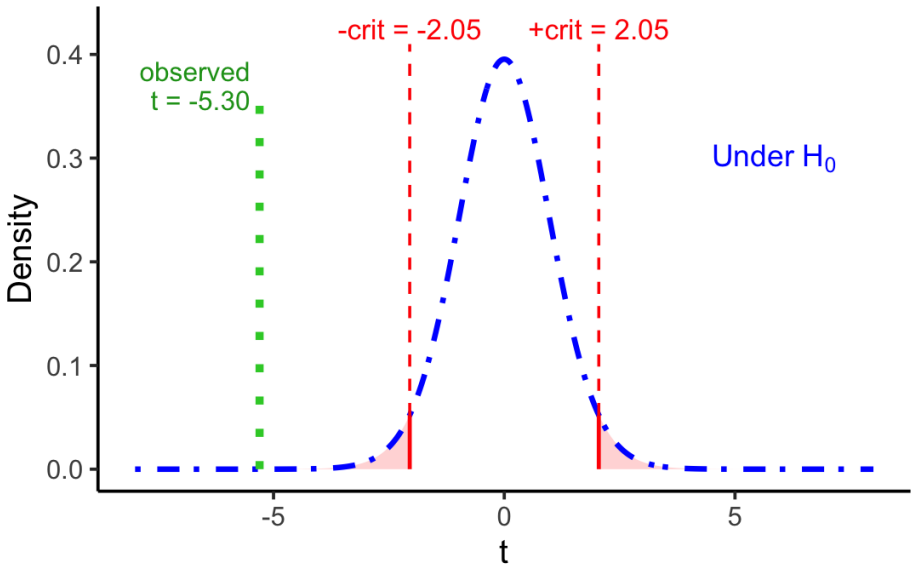


Figure 4: The null distribution of t for the slope test at $df = 28$ (blue dot-dash). The shaded tails beyond the red critical values are the rejection region at $\alpha = 0.05$. The lime-green dotted line marks our observed t, far into the left tail.

a t-distribution with $df = 28$. Figure 4 shows that distribution, with the critical values marking the boundary of the most extreme 5% under H_0, and our observed t marked against it.

Our observed $t = -5.30$ is far past the critical values of about ± 2.05, well outside where 95% of t-values would fall if temperature and DO were unrelated. The p-value is the probability of a t that far out (or farther) in either tail under H_0. Here $p < 0.001$, which crosses the conventional $\alpha = 0.05$ threshold.

We reject H_0 and conclude that stream temperature is linearly related to dissolved oxygen.

The 95% confidence interval for the slope, from `confint(fit)`, in the last two columns of the table above, is 95% CI [-0.50, -0.22] mg/L per °C. Because this interval does not contain 0, it agrees with the hypothesis test, as it always will. The interval also contains $\beta_1 = -0.28$, the true slope we built into the data. A 95% interval catches the true value in about 95% of samples, and this sample is one of them.

Partitioning variation: regression as ANOVA

Besides testing the slope, we want to know how much of the variation in DO the line explains. We answer that the same way as one-way ANOVA in Chapter 8, by splitting the total variation into a piece the model explains and a piece it leaves over.

Figure 5 shows the three pieces for the stream data. Every site's distance from mean DO (left) splits into the distance from the mean to the line (middle), the part explained by the line, and the distance from the line to the point (right), the residual.

Squaring and summing each set of segments gives the partition:

$$SS_{\text{Total}} = SS_{\text{Regression}} + SS_{\text{Residual}}, \qquad 122.44 = 61.35 + 61.10$$

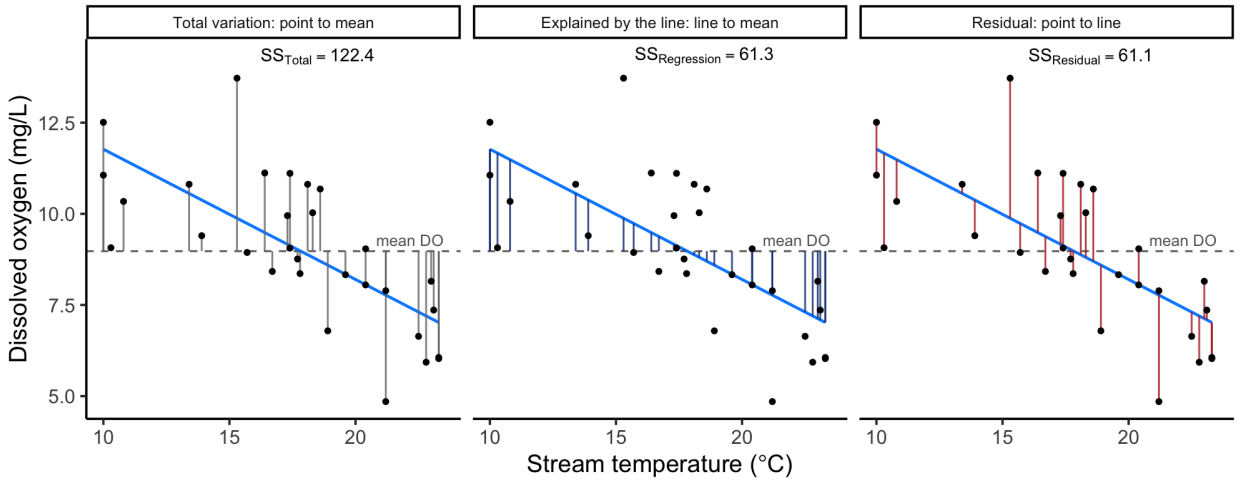


Figure 5: Partitioning variation in dissolved oxygen. Left: each site's distance from mean DO (dashed line). Middle: the part of that distance explained by the line. Right: what is left over, the residuals. Squaring and summing each set of segments gives the three sums of squares.

The pieces line up with Chapter 8's. $SS_{\text{Regression}}$ is the variation the predictor explains, the role SS_{Between} played for group means. SS_{Residual} is what's left over around the line, the role SS_{Within} played around each group's mean. Treating every temperature value as its own group means that the line supplies a predicted mean at each value of x, in place of one mean per group.

R-squared (R^2) is the proportion of the total variation explained by the line:

$$R^2 = \frac{SS_{\text{Regression}}}{SS_{\text{Total}}} = \frac{61.35}{122.44} = 0.50$$

It runs from 0 to 1, and it is the same quantity as η^2 from Chapter 8. For simple regression with one predictor, R^2 is also Pearson's r from Chapter 2, squared: $(-0.708)^2 = 0.501$.

The regression ANOVA table. Dividing each sum of squares by its degrees of freedom gives mean squares, and their ratio is F, exactly as in Chapter 8. The regression row uses 1 degree of freedom, one for the slope. The residual row uses $n - 2$, since the line estimated two things, an intercept and a slope.

```
anova(fit)
```

Table 4: Regression ANOVA table for DO ~ stream temperature.

Source	df	SS	MS	F	p
stream_temp	1	61.35	61.35	28.12	< 0.001
Residuals	28	61.10	2.18		

$$F = \frac{MS_{\text{Regression}}}{MS_{\text{Residual}}} = \frac{61.35}{2.182} = 28.12$$

With one predictor, F and the slope's t test the same thing: $F = t^2 = (-5.30)^2 = 28.12$, with the same p. F is the form that generalizes to models with more than one predictor, which is why software reports both.

Here $R^2 = 0.50$, so stream temperature explains about 50% of the variation we observed in dissolved oxygen across sites. The remaining 50% comes from everything else the model doesn't capture (discharge, shading, groundwater input, and so on), and it shows up as the residuals.

Consider both significance and effect size

As with every test in this book, the significance test is not the whole picture. Testing the slope asks whether β_1 differs from 0. R^2, regression's effect size, says how much of the variation the line explains. The two can disagree in either direction.

A very large sample, the kind common in remote sensing and other big-data environmental work, can make a trivial effect significant, because more observations shrink the standard error until almost any nonzero slope clears $p < 0.05$. A small sample can fail the other way and miss a real, sizeable effect because its standard error is too large. Because p and R^2 answer different questions, we report both.

Figure 6 shows three of these combinations: a strong relationship with a non-significant slope (underpowered), a weak relationship with a significant slope (a large sample), and a relationship that is both strong and significant. In the left panel, six points give $R^2 = 0.63$ but $p = 0.058$. In the middle panel, 5,000 points give $p = 0.004$ for a line that explains 0.2% of the variation. We built a true slope of 0.05 into that panel, so the line rises only half a unit across the whole range of x.

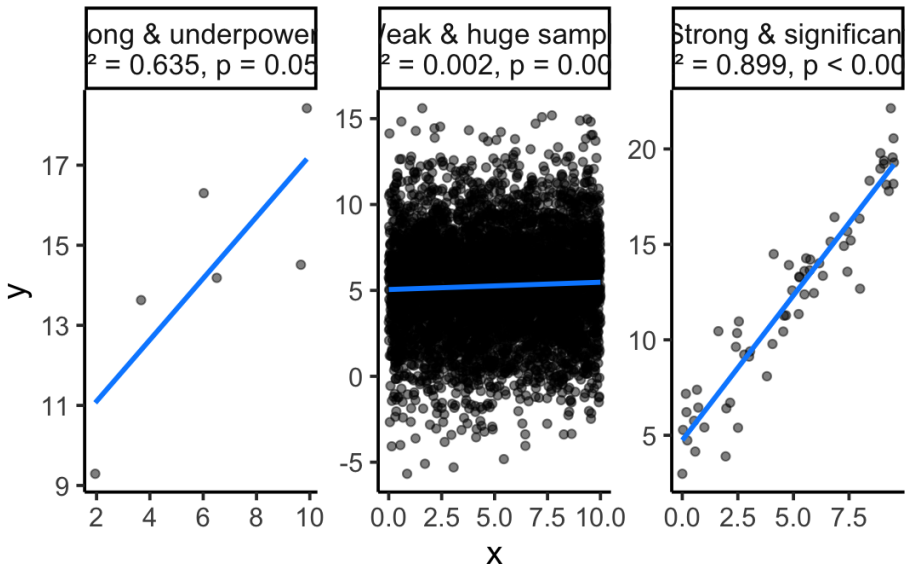


Figure 6: R-squared and the slope's p-value can disagree. Left: a strong relationship (high R-squared) that isn't statistically significant. Middle: a weak relationship (low R-squared) that is significant anyway. Right: strong and significant together.

> Warning: How big is a "good" R^2?
>
> It depends on the field. In studies of human behavior, where many unmeasured factors shape any outcome, 0.20 is a strong result. Environmental systems tend toward more direct physical relationships, so 0.5 to 0.8 for a well-chosen single predictor isn't unusual. An R^2 that looks *too* good, say 0.98 from a single predictor in a noisy field system, more often points to a problem in how the analysis was set up than to a real result.

Checking assumptions

Simple linear regression has three test-specific assumptions, plus the universal independence assumption from Chapter 6. Table 5 lists all four with the check this chapter uses.

Table 5: Assumptions of simple linear regression and how to check them.

Assumption	What it means	How to check	What a violation looks like
Linearity	The true relationship between X and Y is a straight line.	Residuals-versus-fitted plot	A curve, such as a U-shape
Independence	Each observation's error is unrelated to the others'. This is the universal assumption from Chapter 6.	The study design	Sites along the same stream, or repeated measurements over time
Constant variance (homoscedasticity)	The spread of residuals is about the same across the whole range of X, the regression version of Chapter 8's homogeneity of variance.	Residuals-versus-fitted plot	A funnel that widens or narrows
Normal residuals	The residuals are approximately normally distributed.	Q-Q plot and Shapiro-Wilk	Points bending away from the Q-Q diagonal

Sometimes the truth isn't a line at all

> You can do linear regression without thinking about whether the phenomenon you're modeling is actually close to linear. But you shouldn't.
> Jordan Ellenberg, *How Not to Be Wrong*

A straight line is not guaranteed to be the right shape for your data. A residuals-versus-fitted plot checks two of the assumptions above at once. A curve means linearity is violated, and a funnel shape means constant variance is violated.

Figure 7, for our stream temperature data, shows residuals scattered evenly above and below zero, with no curve (which would suggest the true relationship isn't linear) and no funnel shape (which would suggest variance changes systematically across the range of X).

Next, check the residuals for a second data set, 14 days of invented avian influenza case counts. We built the counts from an exponential curve, $6e^{0.28 \times \text{day}}$, plus random scatter, so we know the true shape that a straight line will miss. The design is the same shape as the stream example, with time as the continuous predictor.

A straight line fit to these 14 days gives $R^2 = 0.81$ and $p < 0.001$, which by the benchmark in the "How big is a good R^2?" warning above looks like a solid, significant fit for a field system. Nothing about those numbers alone raises a flag. The plot does raise one, though. The fitted line starts at about -40 cases on day 1, and a count of cases can't be negative.

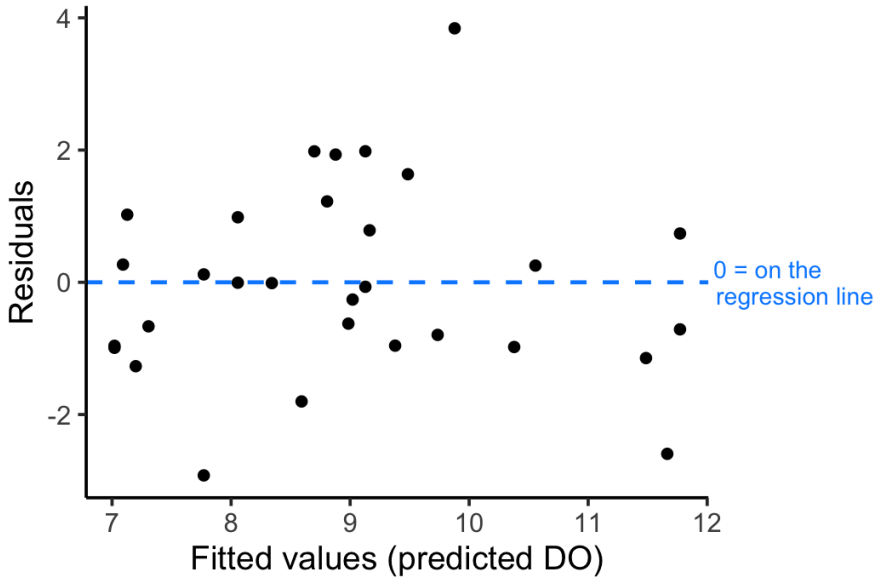


Figure 7: Residuals versus fitted values for the stream temperature model. A random scatter around zero, with no funnel shape or curve, supports linearity and constant variance.

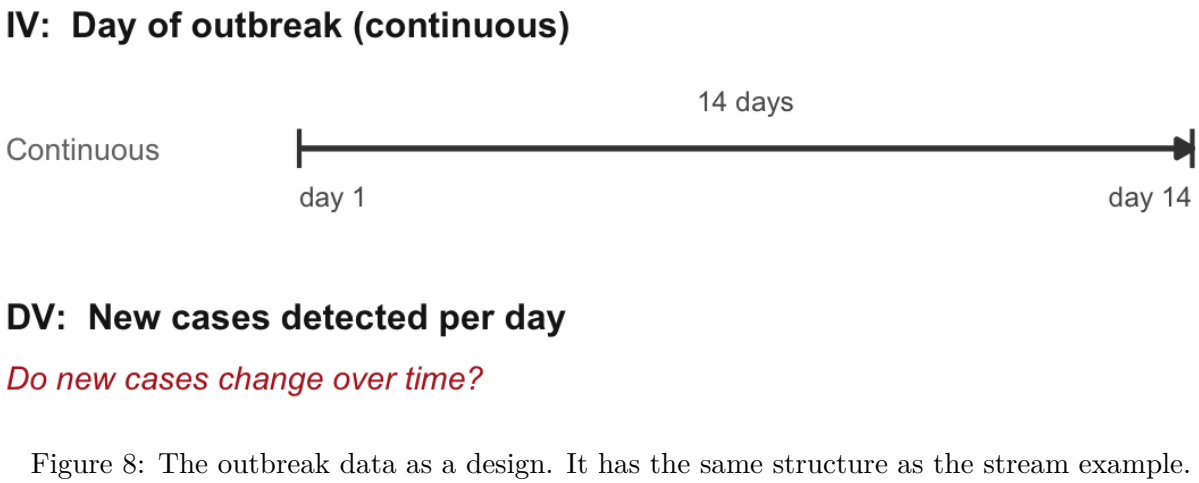


Figure 8: The outbreak data as a design. It has the same structure as the stream example.

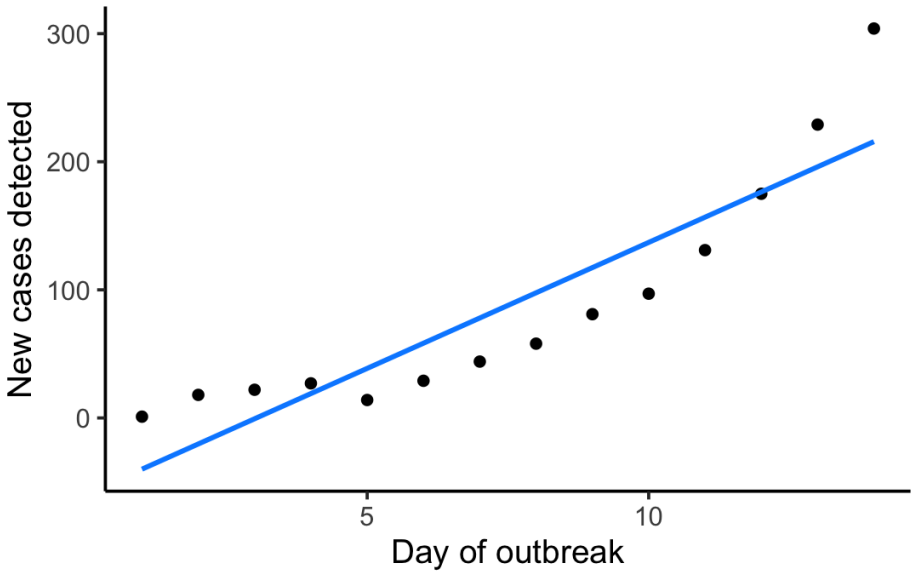


Figure 9: New avian influenza cases per day, days 1-14 of an outbreak, with a fitted straight line.

Check the residuals. The residuals-versus-fitted plot shows a problem that the R^2 and slope test did not.

The residuals form a U. The line underpredicts early and late in the outbreak and overpredicts in the middle, which is the pattern a straight line fit to a curve produces.

If a residual plot curves instead of scattering randomly, a straight line is the wrong model for the data.

Fixes typically include:

- a different model (often adding a polynomial term)
- a transformation, re-expressing a variable on a different scale so a straight line fits it (log-transforming a variable that grows multiplicatively, like the outbreak data above, is the most common case)
- or a nonlinear model entirely, beyond what this course covers

Checking normality

For normality, we check the residuals two ways: visually, with a **Q-Q plot**, and formally, with the Shapiro-Wilk test.

A Q-Q plot sorts the residuals and compares them to where a perfectly normal distribution would place values at the same percentiles. If the residuals are normal, the points fall close to the diagonal line. A systematic curve or S-shape means they don't.

Figure 11 looks close to straight, consistent with normal residuals. Shapiro-Wilk asks the same question formally. Its H_0 is that the residuals come from a normal distribution, so a small p-value is evidence against normality.

```
shapiro.test(resid(fit))
```

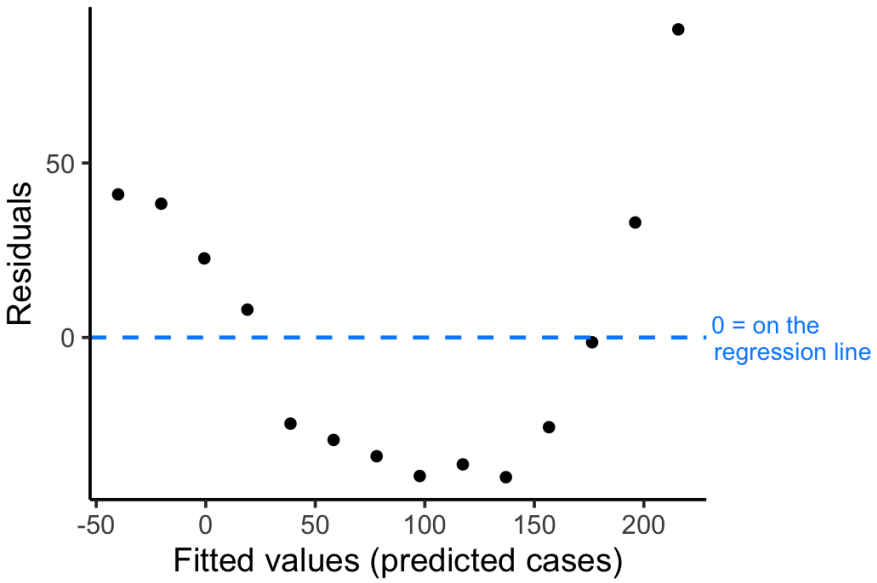


Figure 10: Residuals versus fitted values for the linear outbreak model. A clear U-shape, not random scatter, is the signature of a nonlinear relationship forced into a straight line.

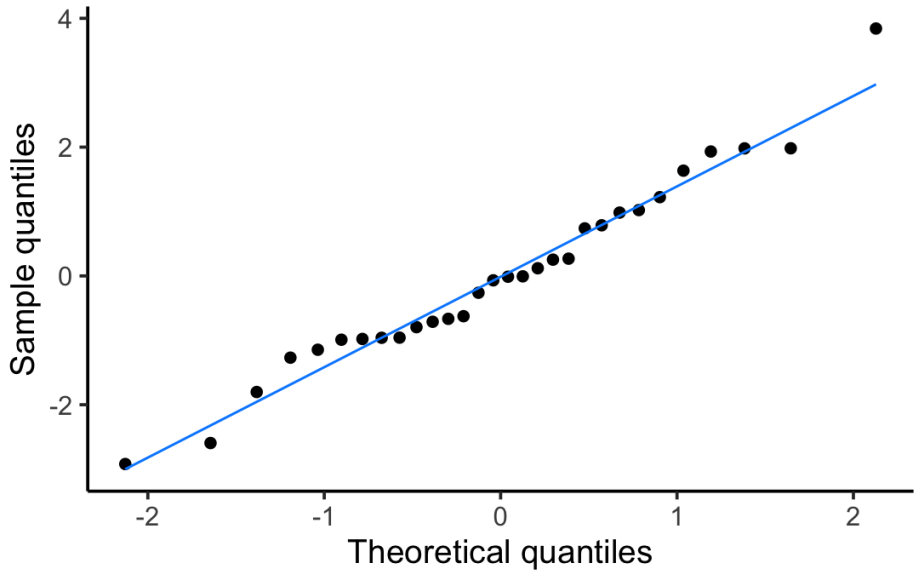


Figure 11: Normal Q-Q plot of the stream temperature model's residuals. Points close to the diagonal support normality.

Table 6: Shapiro-Wilk test on the regression residuals

Statistic	p
0.97	0.663

The residuals don't depart from normality here ($p = 0.663$). As with any formal normality test, treat this as one piece of evidence alongside the Q-Q plot and residual plot, not a strict gate. Shapiro-Wilk has the least power to detect real departures when the sample is small, which is also when the visual checks matter most.

Now apply the same two normality checks to the avian influenza model, the one we already know is badly nonlinear.

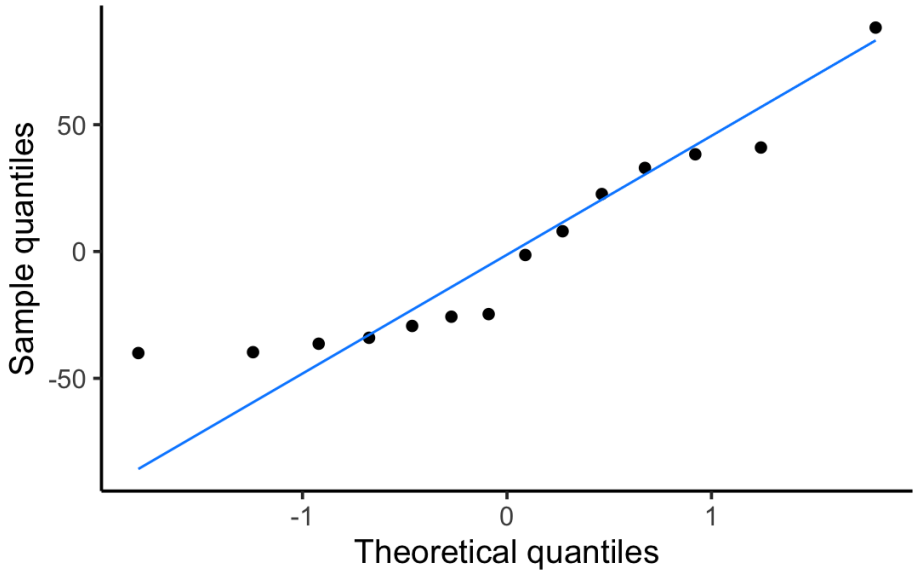


Figure 12: Normal Q-Q plot of the outbreak model's residuals. The lowest residuals bend away from the diagonal, but not by enough for Shapiro-Wilk to reject normality.

The Q-Q plot bends away from the diagonal at the low end, but Shapiro-Wilk does not reject normality ($p = 0.060$). On a model we already know is the wrong shape, the normality checks give at most a borderline signal. They check *normality of the residuals*, which is a different assumption from linearity, and a model can badly violate linearity while its residuals look only mildly non-normal.

! Results reporting: regression

We report a regression result the same way as any other test: the slope, the test statistic with its degrees of freedom, p, the slope's confidence interval, and R^2 as the effect size.

> Dissolved oxygen declined significantly with stream temperature, $b_1 = -0.36$ mg/L per °C, $t(28) = -5.30$, $p < 0.001$, 95% CI [-0.50, -0.22] mg/L per °C, $R^2 = 0.50$.

Using the model: prediction

Once we've confirmed a real linear relationship, the fitted equation lets us predict DO at any stream temperature. Plugging temperature into the fitted line directly gives the prediction: at 20°C,

$$\widehat{DO} = b_0 + b_1(20) = 15.34 + (-0.36)(20) = 8.20$$

R gives the same number, along with a confidence interval. `predict()` takes the fitted model and a small data frame holding the new temperature. The column name, `stream_temp`, has to match the predictor's name in the model.

```
predict(fit, newdata = data.frame(stream_temp = 20), interval = "confidence")
```

Table 7: Predicted dissolved oxygen at 20°C, with 95% confidence interval

Stream temperature	Predicted	CI low	CI high
20	8.2	7.57	8.83

At 20°C, the model predicts DO of 8.20 mg/L, 95% CI [7.57, 8.83]. Because 20°C falls inside the observed range of 10.0 to 23.3°C, this prediction is interpolation, which is what regression is good for. This confidence interval is for the mean DO of all streams at 20°C. A single stream at 20°C will scatter around that mean by about as much as the points scatter around the line, so its DO can fall well outside this interval.

Predicting DO at, say, 35°C would be extrapolation. The fitted line gives 2.8 mg/L at 35°C, a number with no data behind it. Nothing in the data tells us the relationship stays linear (or even stays a relationship at all) beyond the range we actually observed, and the further outside that range, the less trustworthy the prediction. Figure 13 shows both zones on the same line.

Regression is used for extrapolation all the time, and often reasonably. A well-understood physical process can justify projecting a line beyond the data you happened to collect. Extrapolating a straight line is only safe if the true relationship actually stays straight past the range you observed.

The problem is that a narrow window of data can't tell you which kind of curve you're actually on. Figure 14 shows four very different relationships, exponential growth, a sigmoid, a hump-shaped curve, and a truly straight line, each with the same short observed window highlighted in blue. In each panel a straight line (dashed red) is fit to that window and extended. Once you see the full curve (gray), you can see how badly linear extrapolation beyond the range of observed data would perform in each case.

Suppose officials use the linear fitted line from the outbreak example above to project avian influenza cases at day 30, well past the 14 days observed.

In Figure 15, the dashed line keeps rising slowly past day 14, while the true exponential curve rises far above it.

The linear model predicts about 530 cases on day 30. On the exponential curve the data were built from, the count is closer to 26,682, roughly 50 times higher.

Planning surveillance and containment staffing off the linear forecast would leave the response badly underresourced for the outbreak that actually arrives. A straight line can look fine over a short early window and still be very wrong about where that window leads.

This problem calls for a different model. More data or a better-fitting straight line would not fix it. Fitting curves and transformations is beyond this course, but you can recognize when you need one by plotting the data and the residuals.

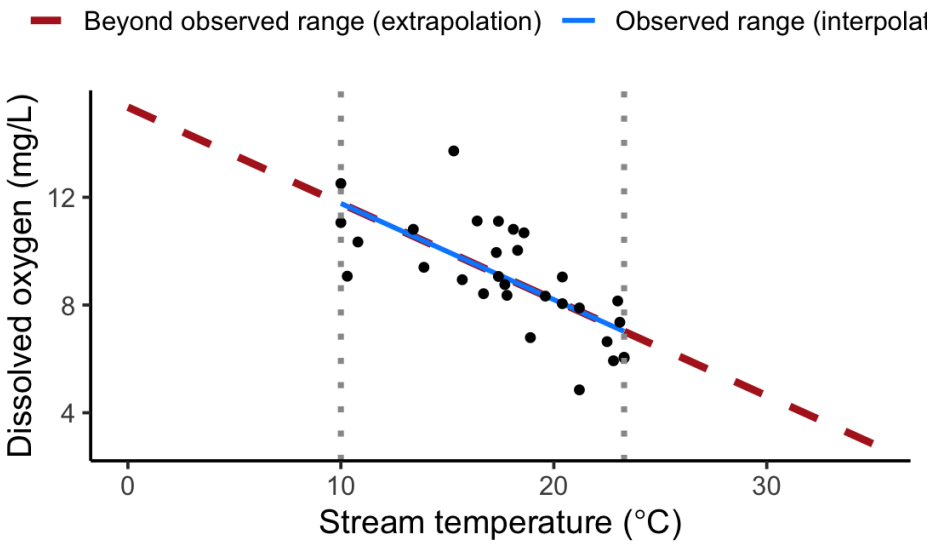


Figure 13: The fitted line within the observed temperature range (solid blue, interpolation) and extended beyond it in both directions (dashed red, extrapolation). Dotted gray lines mark the lowest and highest observed temperatures, 10.0 and 23.3°C.

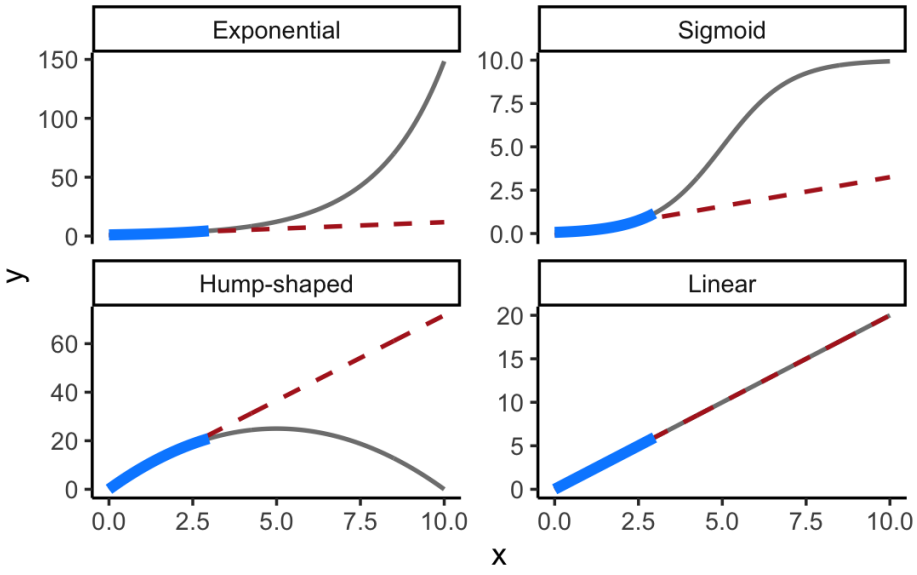


Figure 14: Four different true relationships that all look locally linear over a short observed window (blue). A line fit to each window (dashed red) matches it closely inside the window, but only the linear case stays correct once the full curve (gray) is revealed.

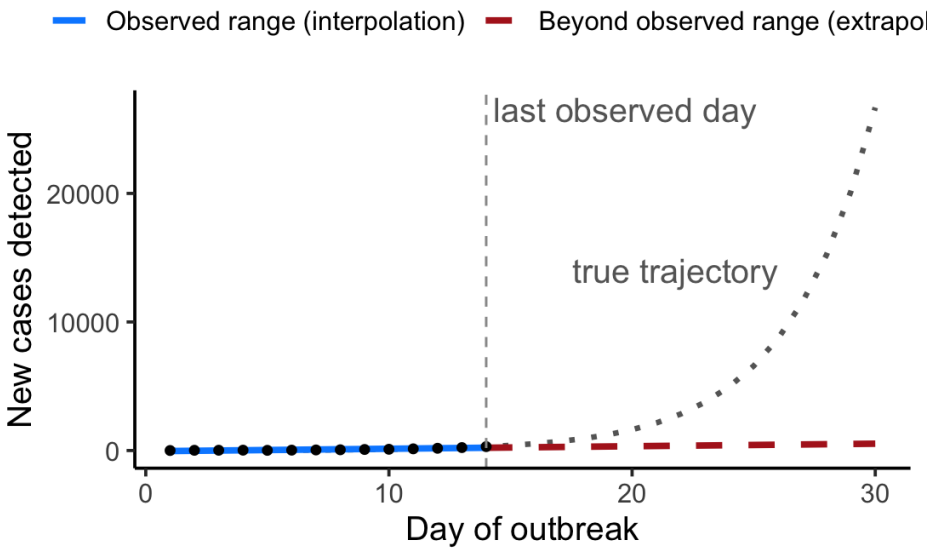


Figure 15: The linear fit extended (dashed) past the observed range, against the true exponential trajectory (dotted gray). The straight line badly underestimates how large the outbreak actually becomes.

Chapter Summary

Why it matters. Chapter 2 drew a best-fit line through a scatterplot. This chapter turns that line into something you can test, quantify, and make predictions from. Chapter 11 adds a categorical factor to this same line.

Core ideas

- **The slope can be tested.** Chapter 2 compared r to a simulated chance window but gave no formal test with a p-value. Fitting a line turns that relationship into a model whose slope we can test: $H_0 : \beta_1 = 0$ says there is no linear relationship. Testing it uses a t-test on the slope divided by its standard error, with $df = n - 2$.
- **Regression partitions variation the way ANOVA does.** SS_{Total} splits into $SS_{\text{Regression}}$ (Chapter 8's between) and SS_{Residual} (Chapter 8's within), and $F = MS_{\text{Regression}}/MS_{\text{Residual}}$, which equals t^2 with one predictor. R^2, the proportion of variation explained by the line, is the effect size. A relationship can be real (p small) and still explain very little, which is why you report both.
- **Look at your data, then look at your residuals.** The test-specific assumptions are linearity, constant variance, and roughly normal residuals, on top of independence. A residuals-versus-fitted plot catches violations a formal test can miss entirely, as the outbreak example showed.
- **Predict inside the observed range, and extrapolate beyond it only with good reason.** Interpolating within the observed range of x is what regression is for. Extrapolating beyond it is sometimes justified, but only when you have good reason to believe the relationship stays the same shape past the range you observed.
- **A good R^2 can still be the wrong model.** The outbreak example fit a straight line to exponential growth and got $R^2 = 0.81$, a residuals-versus-fitted plot that clearly curved, and an extrapolated forecast about 26,000 cases short of the exponential curve the data came from. The residual plot catches this. A normality check on the same residuals gives only a borderline signal, because it tests a different assumption.